# Move to Move?
# Neighborhood Fitness Culture and Exercise Among Older Adults

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# Outline

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# Motivation

- The U.S. is aging rapidly
  - by 2034, adults 65$+$ outnumber children $<18$
- Aging raises concerns about **independence, quality of life, and rising healthcare costs** later in the life
- **Federal Physical Activity Guideline (HHS 2018) :** Adults 65$+$ should do at least :
  - $\geq$ **150 to 300 min/week** of moderate *or*
  - $\geq$ **75 to 150 minutes** of vigorous workout per week
  - $+$ Multicomponent physical activities that include balance strengthening exercise
- **Reality :** Over **25%** of older adults are physically **inactive** *(CDC, 2021)*.
  - Chronic disease, falls, mortality
  - Costs: $\$117$B annually in medical expenditure
- **Policy priority:** Promoting healthy aging through physical activity
  - The role of **Fitness culture**

## Context: Spatial Inequality in Physical Activity

- Older adults' physical activity differs substantially across **regions and communities**
- These **spatial gaps** suggest that physical activity may depend partly on **local environments**
- Older adults do not make exercise decisions in isolation: **local norms, social exposure, and infrastructure may shape behavior**
- If neighborhoods influence exercise, place-based interventions could:
    - Reduce **spatial inequalities** in preventive health,
    - Improve **functional independence**,
    - Lower downstream **medical spending**

# Context: Physical Inactivity

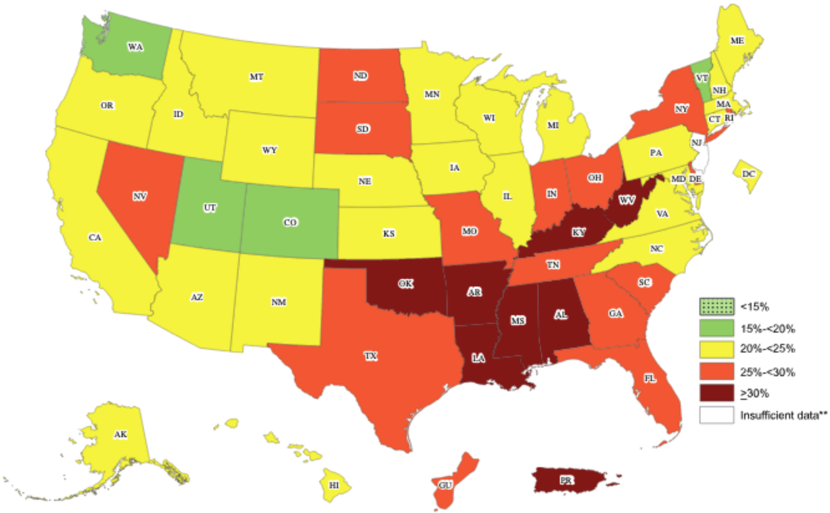


**Figure 1:** Physical inactivity by region – CDC- BRFSS 2020

# Related Literature

- **Neighborhood effects**
  - **Children and economic mobility:** neighborhoods affect long-run outcomes such as income mobility and college attendance (Chetty, Hendren, and Katz, 2016; Chetty and Hendren, 2018).
  - **Older adults' health and mortality:** place-based factors shape mortality and health outcomes among older adults (Finkelstein, Gentzkow, and Williams, 2021; Deryugina and Molitor, 2020, 2021).

- **Physical activity**
  - **Built environment:** walkability, parks, safety, and recreation access are associated with older adults' physical activity (Van Cauwenberg et al., 2011; Barnett et al., 2017).

# Research Gap and This Paper

- The role of **local norms** (**fitness culture**) in shaping older adults' physical activity is largely understudied

**This paper asks:**

*Does moving to a more active neighborhood increase older adults' physical activity?*

- I define **fitness culture** as the local prevalence of **physical activity** among older adults and study whether movers' own behavior **converges toward their new neighborhood fitness culture**

# Outline

# Data

**Health and Retirement Study (HRS), 2004–2022**

- Nationally representative longitudinal survey of older Americans
- Public
  - Self-reported exercise behavior
  - Demographics
  - Health conditions
- Restricted
  - ZIP code level geographic identifier
  - ZIP code level built environment

# Key Variable : Physical Activity Measure

- **Moderate physical activity measure**
  - How often do you take part in sports or activities that are moderately energetic, such as **gardening, cleaning the car, walking at a moderate pace, dancing, stretching exercise**?
- The response is categorical
  - Hardly or never
  - One to three times a month
  - Once a week
  - More than once a week
  - Everyday

# Key Variable : Physical Activity Measure

- Main outcome : **Daily Moderate Physical Activity** - Everyday vs Not
  - Behaviorally meaningful margin : Repeated , habitual physical activity rather than occasional activity
  - Higher chance to meet federal guideline of physical activity ($\geq$ 150 $\sim 300 mins/week$)
  - Moderate activity captures **routine lifestyle behavior embedded in daily life**

# Outline

# Empirical Strategy 1 (following Finkelstein, Gentzkow, and Williams 2021)

**Step 1. Include movers who move only once**

**Step 2. Construct the outcome**

- $\Delta Workout_i = Workout_i^{post} - Workout_i^{pre}$
- Each mover only contributes to **one** row

**Step 3. Construct leave-one-out neighborhood fitness culture**

$$Fitness_{zt}^{(-i)} = \frac{\sum_{j \neq i:\, z_{jt}=z} workout_{jt}}{\sum_{j \neq i:\, z_{jt}=z} 1}$$

**Step 4. Define mover-specific fitness shock**

$$FitnessShock_i^{LOO} = Fitness_{dest(i),\, m_i}^{(-i)} - Fitness_{origin(i),\, m_i - 1}^{(-i)}$$

- $FitnessShock_i^{LOO} > 0$: mover relocates to a more active ZIP.
- The leave-one-out construction excludes mover $i$'s own workout behavior.

# Empirical Strategy 1 (following Finkelstein, Gentzkow, and Williams 2021)

**Step 5. Selection residuals correction**

Regress pre-move workout on observables among movers:

$$Workout_i^{pre} = X_i'\pi + \hat{u}_i$$

Then average the residual $\hat{u}_i$ two ways:

**Origin residual:** $\hat{\hat{u}}_{origin(i)} = \frac{1}{N_o} \sum_{j \in \text{origin}(i)} \hat{u}_j$ *Are people* **from** *this ZIP unobservably fitter?*

**Dest. residual:** $\hat{\hat{u}}_{dest(i)} = \frac{1}{N_d} \sum_{j \in \text{dest}(i)} \hat{u}_j$ *Does this ZIP* **attract** *unobservably fitter movers?*

**Step 6. Main estimating equation**

$$\boxed{\Delta Workout_i = \beta\, FitnessShock_i^{LOO} + X_i'\gamma + \delta_1 \hat{\hat{u}}_{origin(i)} + \delta_2 \hat{\hat{u}}_{dest(i)} + \varepsilon_i}$$

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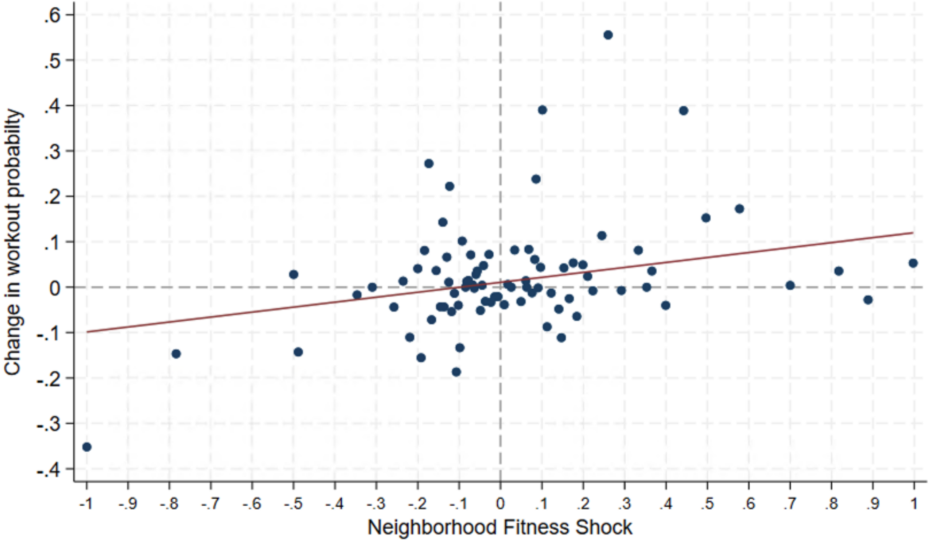
# Visual Evidence



# Results

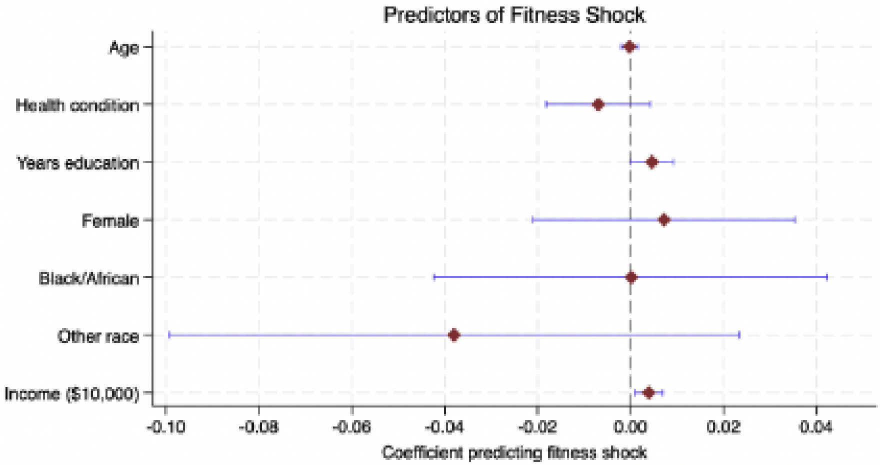
Effect of Fitness Shock on Changes in Moderate Physical Activity

| | Δ Moderate Physical Activity Probability | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Fitness shock | 0.047* | 0.046* | 0.048* |
| | (0.024) | (0.026) | (0.027) |
| Demographic/health controls | No | Yes | Yes |
| Built-environment controls | No | No | Yes |
| Origin/destination selection residuals | Yes | Yes | Yes |
| Mean moderate physical activity | 0.07 | 0.07 | 0.07 |
| Observations | 1,419 | 1,408 | 1,388 |

Notes: The dependent variable is the change in moderate physical activity, Δ Moderate Physical Activity. Fitness shock is measured using the leave-one-out neighborhood fitness shock. All specifications include origin and destination selection residuals. Column (1) reports the baseline specification. Column (2) adds individual demographic and health controls, including age, years of education, gender, race, and health condition (Hypertension, diabetes, cancer, lung disease, heart disease, stroke, psychiatric problems and arthritis). Column (3) further adds built-environment controls, including total park area and recreation-venue density. Standard errors are reported in parentheses. Column (1) clusters standard errors at the individual level; Columns (2)–(3) cluster standard errors at the origin ZIP level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

# Observables predict fitness shock?




*Joint F-test of covariates:* $F(1, 1406) = 6.60$, $p = 0.0103$.

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# Empirical Strategy 2: Motivation

**Concern with the mover design:** movers may select destinations based on **unobserved fitness preferences**, even after the selection residual corrections.

**Solution:** restrict to movers whose reason for moving is plausibly *exogenous* to fitness culture

- HRS collects reason for move **( But most of them are missing )**
- Natural disaster (35) , Foreclosure (26) , **Weather and climate ( 191 )**

**Control group:** never-movers observed over the same period.

**Design:** event study comparing workout trajectories of **weather-induced movers vs. never-movers**

## Empirical Strategy 2: Identification

- Move is driven by climate preferences $\Rightarrow$ destination choice is plausibly orthogonal to fitness culture.
- Any fitness culture shock they experience may be *incidental* to the move decision.
- If the coefficient reflected **selection (fitter people choose places with better weather)**, we would expect a positive or null effect for *both* up- and down-movers.
  - **Up-movers:** climate-induced movers who move to more active area
  - **Down-movers:** climate-induced movers who move to less active area

# Empirical Strategy 2: Setup

**Leave one out Fitness shock at the move:**

$$FitnessShock_i^{LOO} = Fitness_{dest(i),\, m_i}^{(-i)} - Fitness_{origin(i),\, m_i - 1}^{(-i)}$$

**Event study specification:**

$$Workout_{it} = \sum_{r \neq -1} \beta_r \left( \mathbf{1}[t - m_i = r] \times FitnessShock_i \right) + X_{it}'\gamma + \alpha_i + \mu_t + \varepsilon_{it}$$

- $r \in \{-4, -3, \dots, -2, 0, 1, \dots, 4\}$; $r = -1$ is the reference period
- $\alpha_i$: individual fixed effects; $\mu_t$: year fixed effects
- SEs clustered at the individual level

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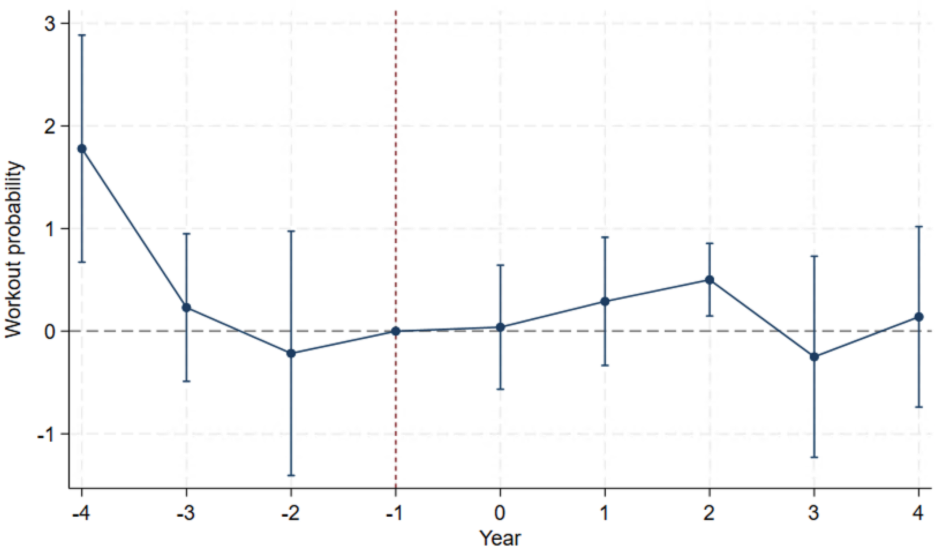
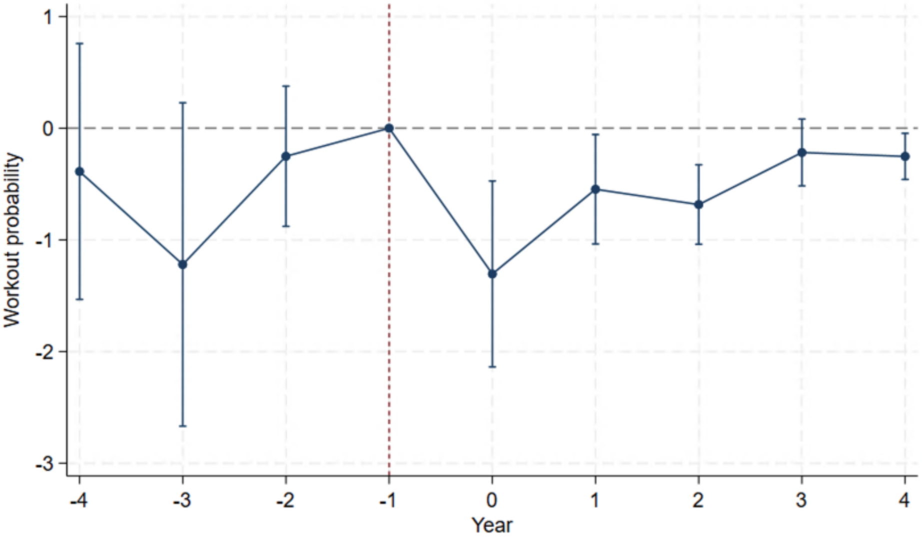
# Event Study - Up mover



# Event Study - Down mover



# Interpretation

- Finding a *negative* effect for down-movers rules out selection: people who moved for weather did not choose less fitness—**They were incidentally exposed to it**
- Asymmetry (larger down-mover effect) is consistent with social/cultural transmission - **Easier to lose a fitness habit than to gain one**

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# Conclusion & Policy Implications

- The results suggest that **physical activity** is shaped not only by individual choice, but also by **neighborhood fitness cultural context** - Visibility matters
- Investments in **age-friendly community exercise opportunities** may help increase exercise participation among older adults
- Such policies may be especially important in neighborhoods where **opportunities for healthy aging are currently limited**